Qiskit 101

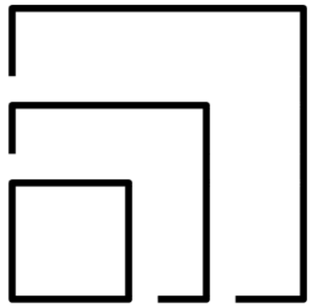
Brief introduction to Qiskit and its main functionalities



Afonso Azenha
University of Stuttgart
Qiskit Fall Fest @ University of Stuttgart
(Slides inspired by **Sebastian Brandhofer's** talk from last year)

IBM

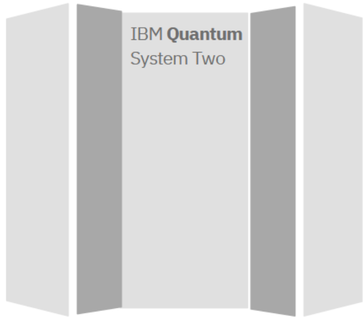
Quantum Software? **Qiskit**? What are those? Why do we need them?



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Quantum Software
(Powered by Qiskit Addons and Qiskit Functions)

Qiskit
(Quantum SDK)

QPUs
(Quantum Processing Unit)

Useful Work

The **IBM Quantum platform** unlocks research with quantum

Qiskit Pattern:
The anatomy of a quantum algorithm



01
Map problem instance to quantum circuits and operators

02
Optimize for target hardware execution

03
Execute via Qiskit Runtime

04
Result processing

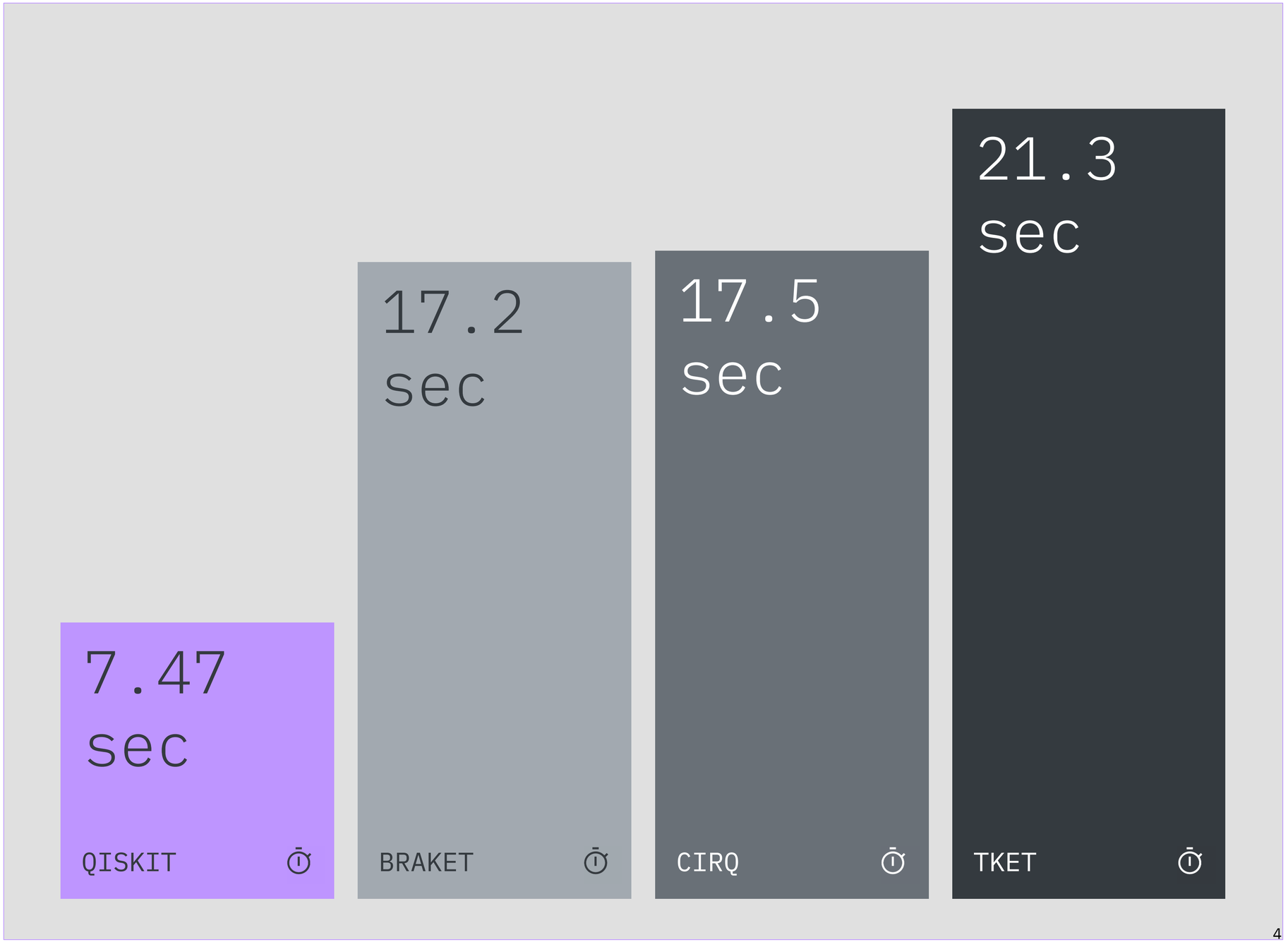
Map

Optimize

Execute

Post-Process

Speed:
Build & Manipulate Time to Complete



Not included due to insufficient test results: CUDA & Staq

*Timing measured using only completed tests

Data based on arXiv:2409.08844 paper published 2024.SEP.13

Quality:
Ratio of 2Q Gates

Quality measures the number of two-qubit gates in a transpiled circuit. Fewer operations means less noise and better results.



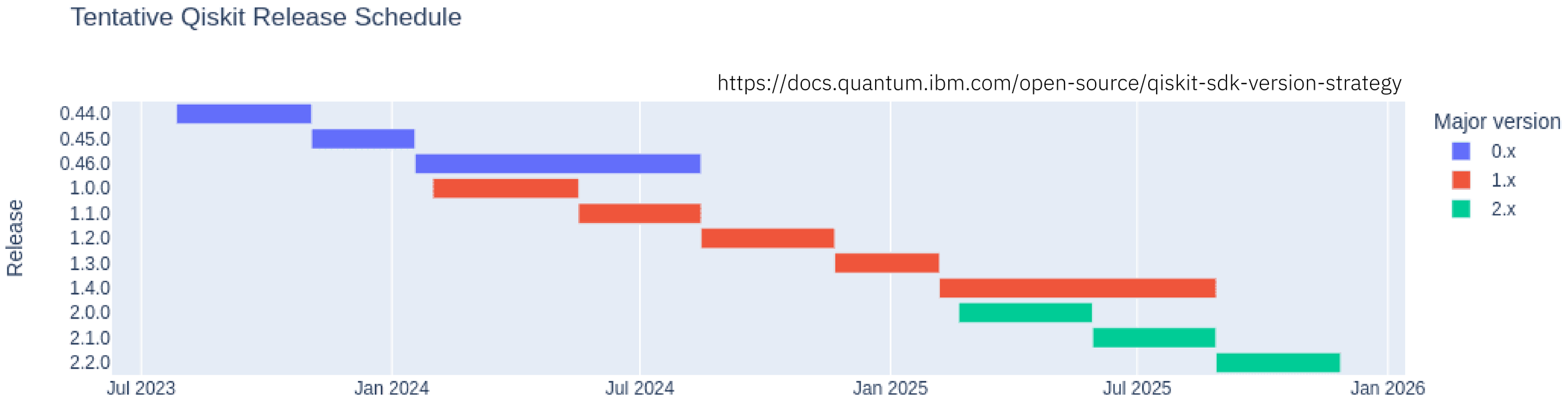
As of 2024.SEP.16

Not included due to insufficient test results: Braket, Cirq & CUDA

*Quality measured using only completed tests

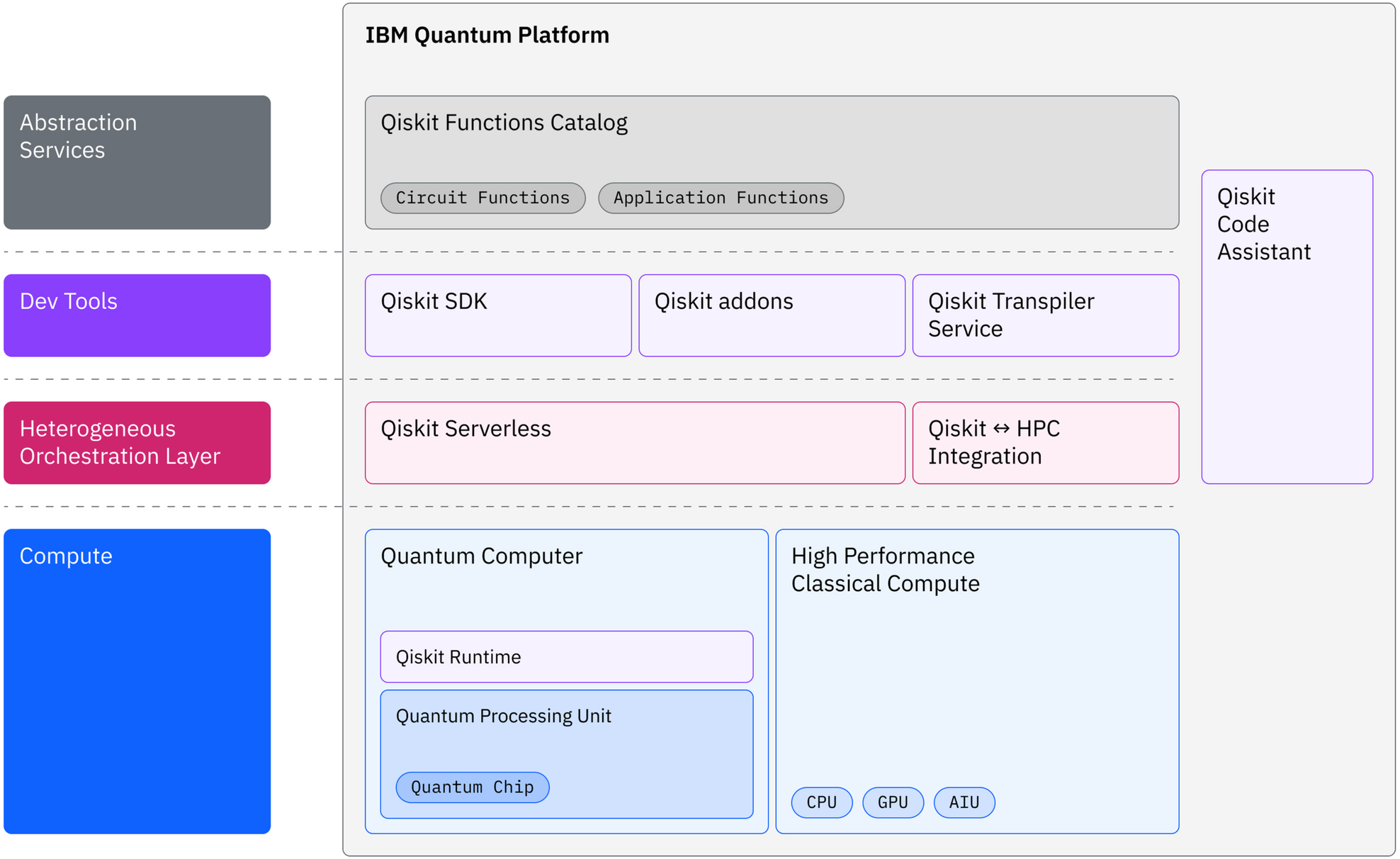
Data based on arXiv:2409.08844 paper published 2024.SEP.13

Qiskit Stability Policy



The Qiskit SDK **guarantees backwards compatibility and bugfixes for at least 1.5 years**

The IBM Quantum Platform



Qiskit Hands-on!



If you've joined the discord server, the notebook has been provided there!